

Understanding Evaluation and Model Performance in Machine Learning

We already learned:

Data → Regression → Linear Regression →
Cost Function → Gradient Descent →
Prediction

But building a model is not the end.

Now we ask: Did the model learn correctly?

This stage is called **Evaluation**.

What is Evaluation?

Evaluation means: Measuring how well a machine learning model performs.

A model may make predictions, but are those predictions:

- accurate?

- reliable?
- useful?

Evaluation answers this.

Logic

Imagine a student gives an exam.

Studying alone is not enough.

We must check:

What score did the student get?

Similarly:

Training a model is not enough.

We must evaluate it.

Purpose of Evaluation

Evaluation helps us:

- measure performance
- compare models

- detect problems
- improve prediction quality
- **Connection**
- Model Training
- ↓
- Prediction
- ↓
- Need quality checking
- ↓
- **Evaluation**

What are Performance Metrics?

Evaluation needs numbers. These numerical measures are called Performance Metrics

Different problems need different metrics.

This is important.

There is no single metric for everything.

Big Idea

Problem Type



Choose an Appropriate Metric

Since we studied Regression, we start with Regression Metrics.

Regression Performance Metrics

Regression predicts continuous values.

Examples:

- salary
- price
- temperature

So we measure:

How far is prediction from reality?

A) MAE — Mean Absolute Error

Formula:

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of samples
- **Meaning**
- MAE calculates:
- Average absolute error.
- Example:
- Actual: 100
- Prediction: 90
- Error: 10

- Absolute means: No negative sign.
- **Logic**
- MAE asks:
 - On average, how wrong are predictions?

Why Use MAE?

Advantages:

- easy to understand
- robust to outliers
- same unit as target

Where Used?

Useful when:

- all errors matter equally

Examples:

- temperature prediction

- sales prediction
- **B) MSE — Mean Squared Error**
- Formula:
- $MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$
- This is also our **cost function** in linear regression.
- ---
- **Meaning**
- Instead of absolute error, We square errors.

Logic

MSE asks:

How large are prediction errors, especially big mistakes?

Large errors receive bigger punishment.

Example:

Error 2:

$$2^2 = 4$$

Error 10:

$$10^2 = 100$$

Huge difference.

Why Use MSE?

Advantages:

- strongly penalizes large mistakes
 - mathematically smooth
 - works well with gradient descent
-

Where Used?

When:

Large errors are dangerous.

Examples:

- medical prediction
- financial forecasting

C) RMSE — Root Mean Squared Error

Formula:

$$RMSE = \sqrt{MSE}$$

or

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

Meaning

RMSE is:

Square root of MSE.

Why?

Because MSE squares units.

Example:

Price:

rupees

MSE:

rupees²

Hard to interpret.

RMSE converts back.

Logic

RMSE asks:

What is the typical prediction error in original units?

Why Use?

- . interpretable
 - . keeps large-error penalty
 - . easier explanation
-

Where Used?

Very common in:

- . forecasting
- . regression projects
- . Kaggle competitions

D) R² Score (Coefficient of Determination)

Now something different.

Previous metrics measured error.

R² measures:

How much variance the model explains.

Formula:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Where:

. $\bar{y}$ =mean value

Meaning

R² compares:

Model

VS

Simply predicting average.

Range

Usually:

$0 \rightarrow 1$

Example

$R^2 = .90$

Means:

Model explains 90% of variation.

Logic

R^2 asks:

How well does the model understand data patterns?

Why Use?

Good for:

- comparing regression models
 - understanding explanatory power
-

Limitation

High R^2 :

Does NOT always mean perfect model.

Overfitting can still happen.

Important point.

Metrics Connection

Evaluation



Need measurement



Performance Metrics



Regression Metrics



MAE → average error

MSE → penalize big errors

RMSE → interpretable error

R^2 → explained variance

What is Overfitting?

Now comes a major problem. Suppose the model performs amazingly on the training data, but poorly on new data. This is Overfitting.

Logic

Imagine a student memorizes answers instead of understanding concepts.

Exam changes student struggles. That is overfitting.

Signs

Training accuracy: very high

Testing accuracy: low

Why Happens?

Usually:

- model too complex

- too many features
 - too much memorization
 - insufficient data
-

Problem: Poor generalization.

Meaning: Fails on unseen data.

What is Underfitting?

Opposite problem.

Definition:

The model is too simple and fails to learn important patterns.

Logic

The student did not study enough. Performs badly everywhere.

Signs

Training: low

Testing: low

Causes

- oversimplified model
- insufficient learning
- poor features

Overfitting vs Underfitting

Underfitting

Too simple

Learns too little

Poor train + test

Overfitting

Too complex

Learns too much

Great train, poor
test

Connection

Model Evaluation



Sometimes poor performance



Need diagnosis



Underfitting or Overfitting

Ridge and Lasso Regression

Now, how do we fix overfitting?

Answer: Regularization

Regularization:

Technique to reduce overfitting.

Two famous methods:

- Ridge
- Lasso

Ridge Regression (L2)

Ridge modifies cost function.

Original:

$$MSE$$

Ridge adds penalty:

$$Cost = MSE + \lambda \sum w^2$$

Where:

- w =weights
- λ =regularization strength

Meaning

Large coefficients get penalized.

Logic

Ridge says:

"Don't let weights become too extreme."

This keeps the model smoother.

Why Use?

- reduces overfitting
 - handles multicollinearity
 - stabilizes model
-

Important Point

Ridge:

reduces coefficients, but rarely makes them exactly zero. So, Features stay, just smaller.

Lasso Regression (L1)

Lasso uses an absolute penalty.

Formula:

$$Cost = MSE + \lambda \sum |w|$$

Meaning, Penalty based on absolute values.

Logic

Lasso says:

"Unimportant features should disappear."

Why Special?

Lasso can make:

coefficients = 0

Meaning:

automatic feature selection.

Where Used?

When:

- many features exist
- some are irrelevant

Ridge vs Lasso

Ridge

Lasso

L2 penalty

L1 penalty

Squares weights

Absolute weights

Shrinks

Can remove coefficients

Ridge

Lasso

coefficients

Keeps all features

Performs feature
selection

Final Full Connection

We build regression model



Need checking



Evaluation



Need numerical measurement



Performance Metrics



MAE / MSE / RMSE / R^2



Metrics reveal model quality



Sometimes model performs poorly



Underfitting or Overfitting



Overfitting needs control



Regularization



Two major methods



Ridge (shrink)

Lasso (shrink + select)

After building a regression model, evaluation measures performance using metrics like MAE, MSE, RMSE, and R^2 . These metrics help identify whether a model is underfitting or overfitting, and if overfitting occurs, regularization techniques such as Ridge and Lasso regression help improve generalization and model stability.